

BLUESTOCK MUTUAL FUND ANALYTICS

Final Project Report

ETL • Exploratory Data Analysis • Fund Performance • Dashboard • Advanced Risk
Analytics

Prepared from the supplied processed datasets and completed analytical outputs

1. Executive Summary

This project builds an end-to-end mutual-fund analytics workflow covering data ingestion, cleaning, SQL storage, exploratory analysis, fund performance measurement, Power BI dashboarding, and advanced risk/investor analytics.

The supplied analytical universe contains 40 schemes, 46,000 NAV observations, 32,778 investor transactions, and benchmark/index observations across NIFTY 50 and NIFTY 100.

The supplied SIP dataset reaches 31,002 crore in December 2025, while total folios increase from 13.26 crore in January 2022 to 26.12 crore in December 2025.

Performance analytics ranks funds using the specified 3-year return, Sharpe, Alpha, expense-ratio inverse, and maximum-drawdown inverse components. The resulting scorecard is used as the basis for dashboard ranking.

2. Data Sources and Scope

The project uses the cleaned CSV datasets: fund master, NAV history, AUM by fund house, monthly SIP inflows, category inflows, industry folio counts, scheme performance, investor transactions, portfolio holdings, and benchmark indices.

The NAV history covers the supplied project period beginning in January 2022. The supplied dataset does not provide a full five years through the analysis end date, so 5-year CAGR remains unavailable where insufficient history exists rather than being fabricated.

Industry KPI values requested by the dashboard brief (₹81L Cr AUM, ₹31K Cr SIP inflows, 26.12 Cr folios, and 1,908 schemes) are treated as assignment reference snapshots. Scheme/fund-house charts are based on the supplied project datasets.

3. ETL and Data Quality

The ingestion scripts load the raw CSV files, standardize column names, remove duplicate rows and parse date fields. Dataset-specific checks cover NAV positivity, transaction-type standardization, transaction amount validation, and scheme-performance validation.

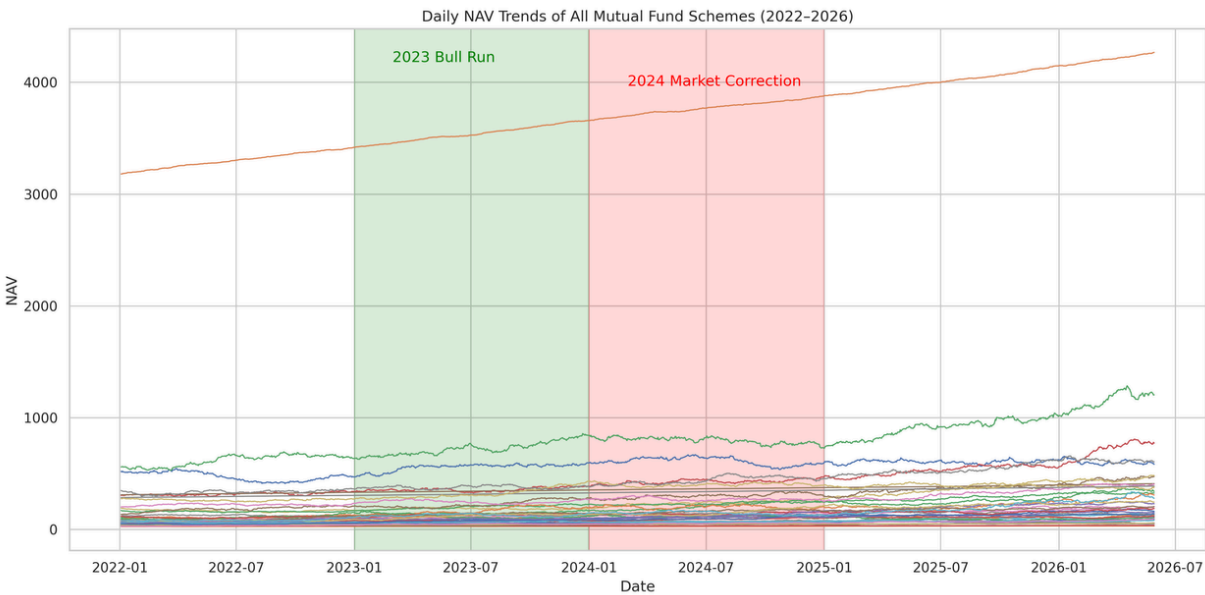
The SQLite layer stores the processed datasets and supports analytical SQL queries. AMFI validation checks the scheme-code relationship between the fund master and NAV history.

An important limitation in the inherited cleaning implementation is that forward-filling without reindexing does not itself create missing weekend/holiday rows. This is documented as a pipeline improvement rather than hidden.

4. EDA — NAV and AUM

NAV trend analysis provides daily movement for all 40 schemes. AUM analysis shows the supplied fund-house-level AUM distribution and its year-level movement.

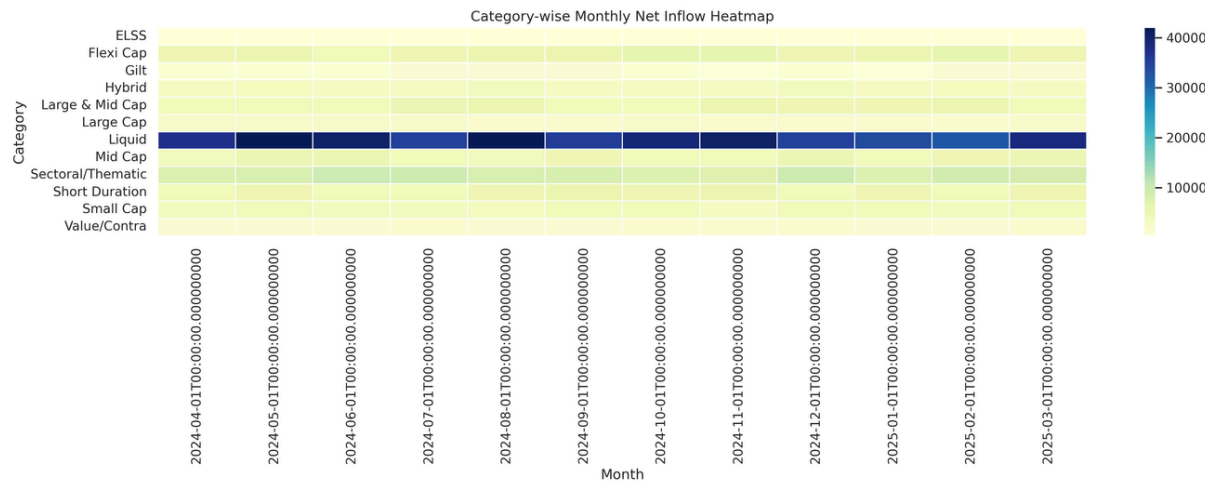
In the supplied AUM dataset, annual grouped totals are 2022: 61.08 Lakh Cr, 2023: 69.77 Lakh Cr, 2024: 143.70 Lakh Cr, 2025: 117.21 Lakh Cr. SBI Mutual Fund is the largest fund-house snapshot at ₹12.5 Lakh Cr in the supplied data.



5. EDA — SIP and Category Flows

Monthly SIP inflows peak at ₹31,002 crore in December 2025. The processed category dataset supports month-by-category flow analysis and the required FY25 ranking.

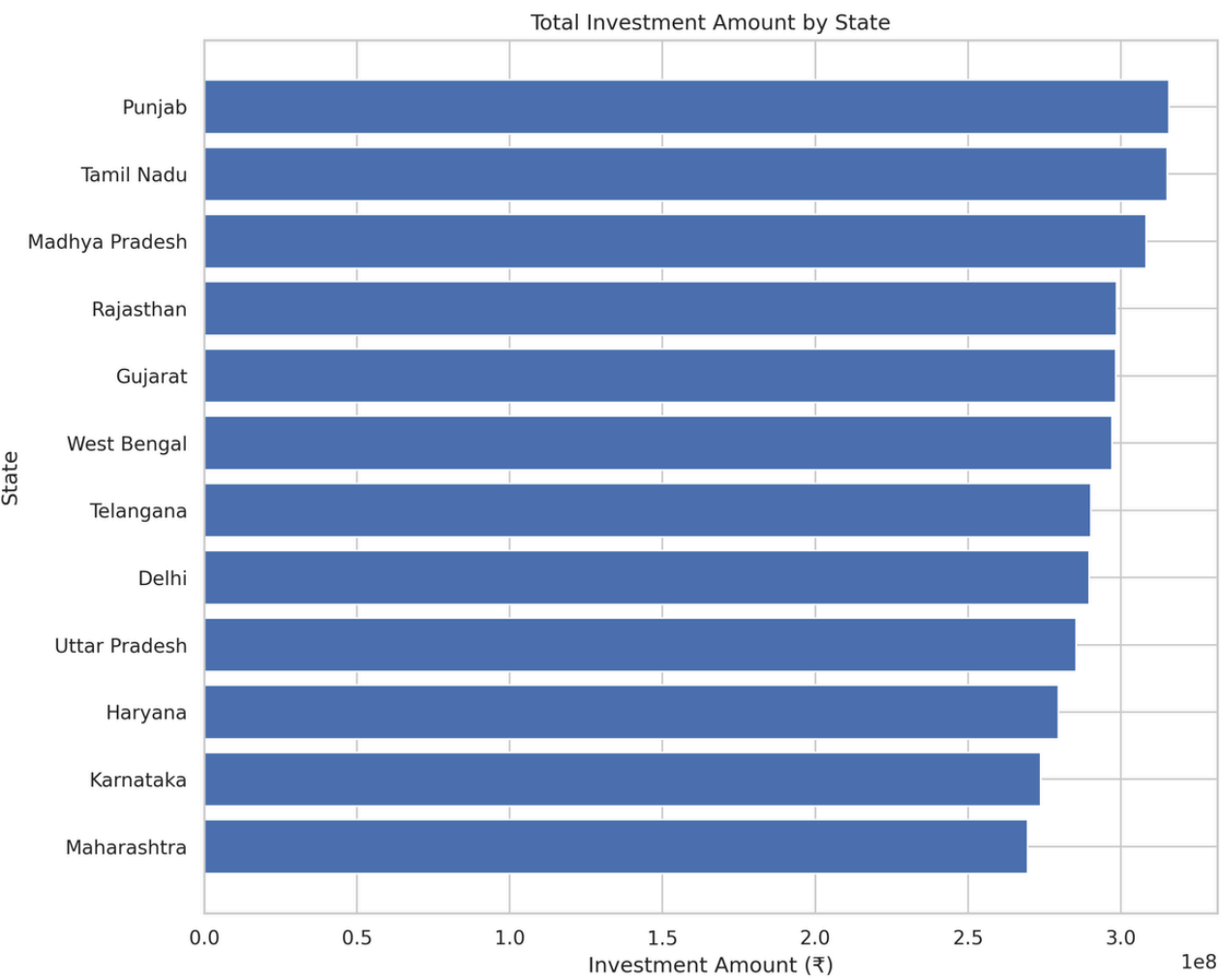
The dashboard and report use the fiscal-year definition April 2024 through March 2025 for FY25 rather than treating calendar 2025 as FY25.



6. EDA — Investor Demographics and Geography

Investor transactions are analyzed by age group, gender, state and city tier. The required dashboard view uses transaction amount by state and separates SIP, Lumpsum and Redemption activity.

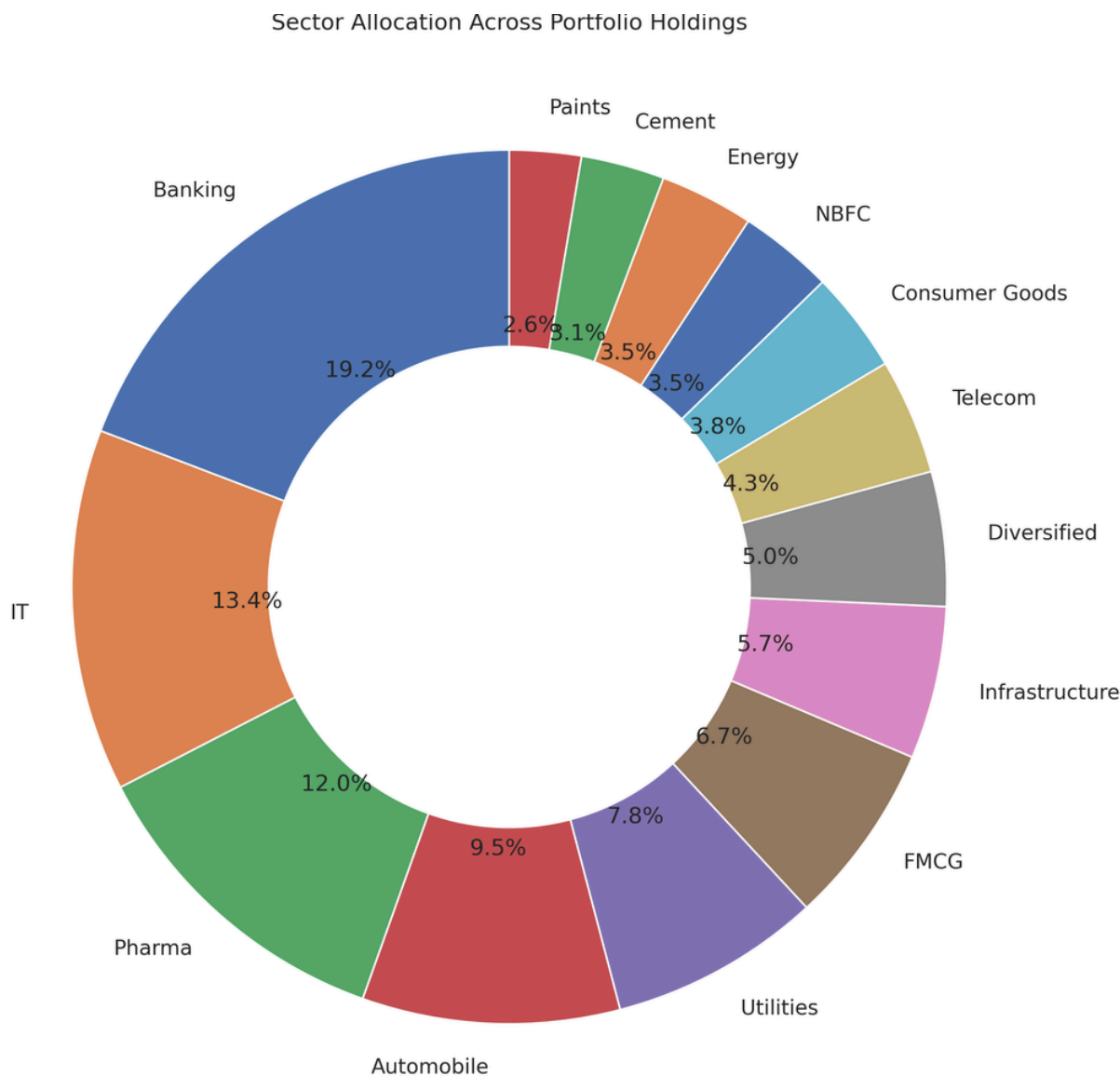
The supplied transaction data contains both T30 and B30 city-tier observations, allowing the required city-tier comparison.



7. EDA — Folios, Correlation and Sector Allocation

Total folios rise from 13.26 crore in January 2022 to 26.12 crore in December 2025.

Return correlation and sector allocation provide portfolio-level context. Sector HHI is additionally calculated in the advanced analytics stage for equity-fund concentration.



8. Fund Performance Methodology

Daily returns are calculated as NAV ending / NAV beginning – 1. CAGR uses the beginning and ending NAV over the relevant period. Sharpe uses a 6.5% annual risk-free proxy and 252 trading days; Sortino replaces total volatility with downside volatility.

Alpha and Beta are estimated by OLS regression of fund daily returns against NIFTY 100 daily returns using `scipy.stats.linregress`. Alpha is annualized by multiplying the regression intercept by 252.

Maximum drawdown is calculated from the running NAV maximum. Tracking error is the annualized standard deviation of the fund-minus-benchmark daily return difference over the three-year comparison window.

9. Fund Scorecard Results

The score card applies the assignment weighting: 30% 3-year return rank, 25% Sharpe rank, 20% Alpha rank, 15% inverse expense-ratio rank and 10% inverse maximum-drawdown rank.

The top-ranked supplied fund is **ICICI Pru Midcap Fund - Regular - Growth** with a score of **84.50**. The ranking is a composite decision-support score, not an investment recommendation.

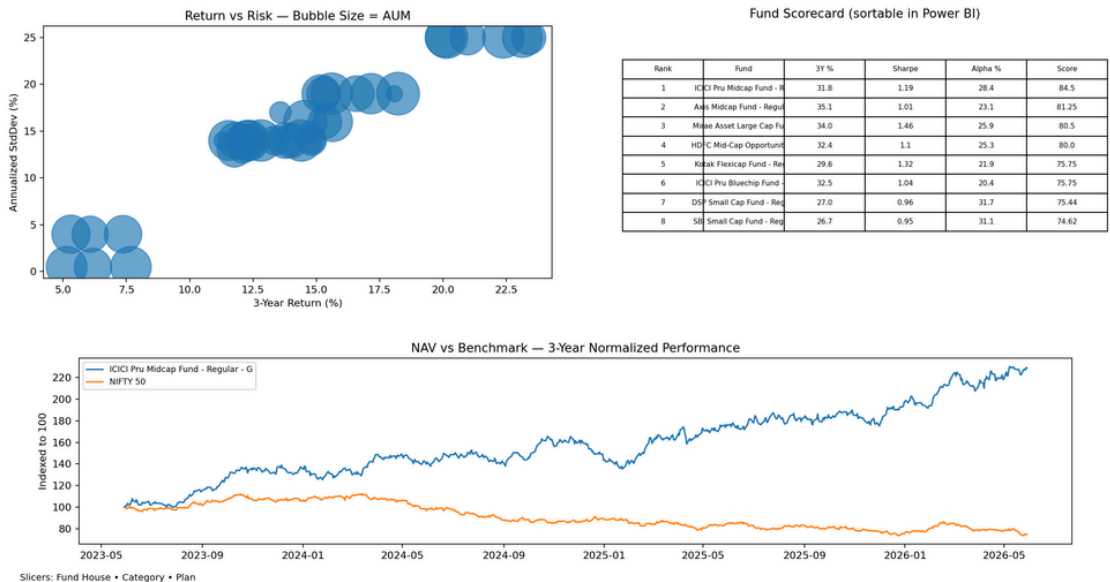
Rank	Fund	3Y CAGR	Sharpe	Alpha	MDD	Score
1	ICICI Pru Midcap Fund - Regula	31.8%	1.19	28.4%	-18.2%	84.50
2	Axis Midcap Fund - Regular - G	35.1%	1.01	23.1%	-21.0%	81.25
3	Mirae Asset Large Cap Fund - R	34.0%	1.46	25.9%	-11.3%	80.50
4	HDFC Mid-Cap Opportunities Fun	32.4%	1.10	25.3%	-16.2%	80.00
5	Kotak Flexicap Fund - Regular	29.6%	1.32	21.9%	-13.0%	75.75
6	ICICI Pru Bluechip Fund - Dire	32.5%	1.04	20.4%	-12.6%	75.75
7	DSP Small Cap Fund - Regular -	27.0%	0.96	31.7%	-31.2%	75.44
8	SBI Small Cap Fund - Regular P	26.7%	0.95	31.1%	-28.7%	74.62

10. Benchmark and Risk Comparison

The benchmark comparison uses NIFTY50 and NIFTY 100. The dashboard design compares the selected fund NAV with a benchmark over the latest three-year window, while Alpha/Beta uses NIFTY 100 as specified.

Fund-level tracking error is reported in the performance outputs and is aligned to the same three-year comparison window for dashboard consistency.

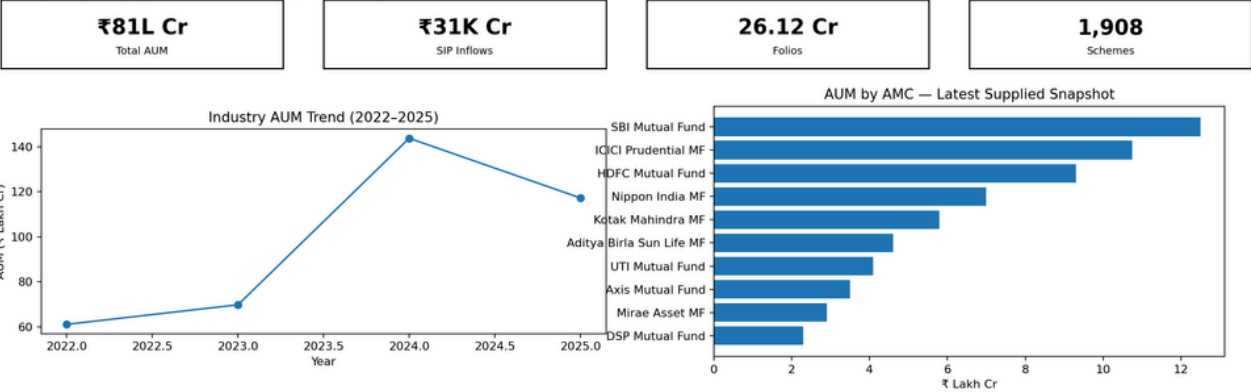
BLUESTOCK — FUND PERFORMANCE



11. Dashboard — Industry Overview

The IndustryOverview page contains the four assignment KPI cards, an AUM trend and AUM-by-AMC view. The KPI cards use the assignment reference snapshot; supporting trend and AMC visuals use the supplied processed datasets.

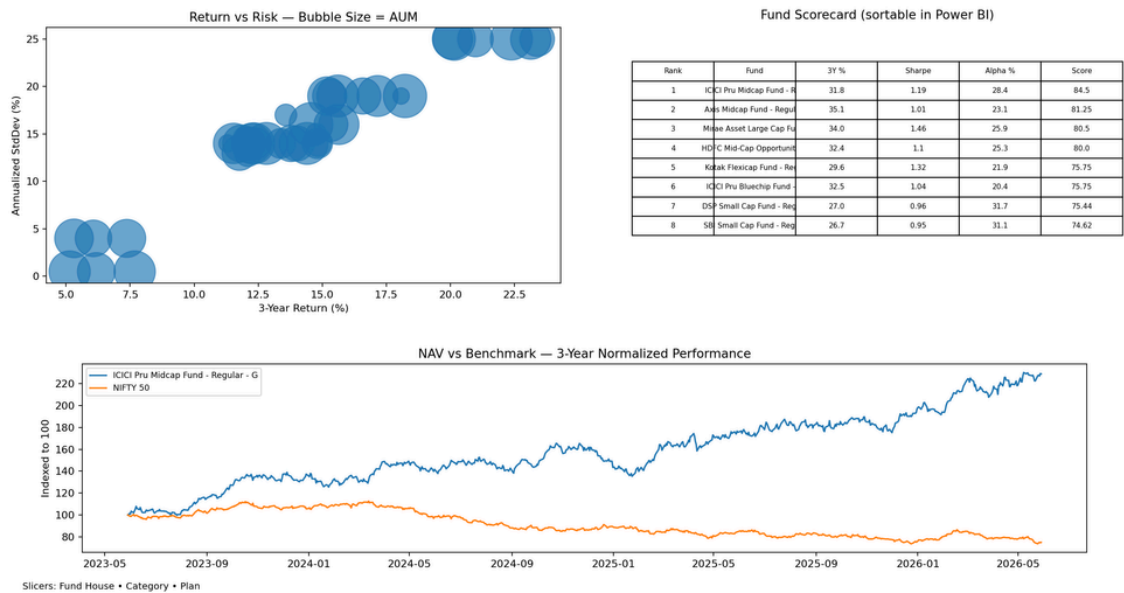
BLUESTOCK — INDUSTRY OVERVIEW



12. Dashboard — Fund Performance

The required fund-performance page uses a return-versus-risk scatter with AUM as bubble size, a sortable score card, a NAV-versus-benchmark view, and slicers for fund house, category and plan. In the native PBIX these elements should remain interactive.

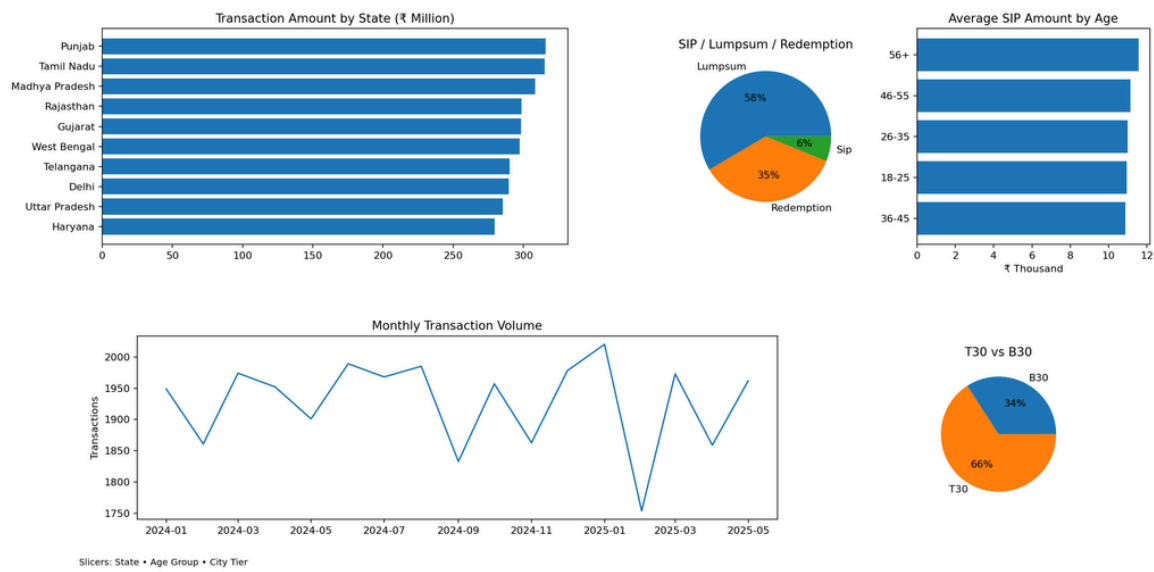
BLUESTOCK — FUND PERFORMANCE



13. Dashboard — Investor Analytics

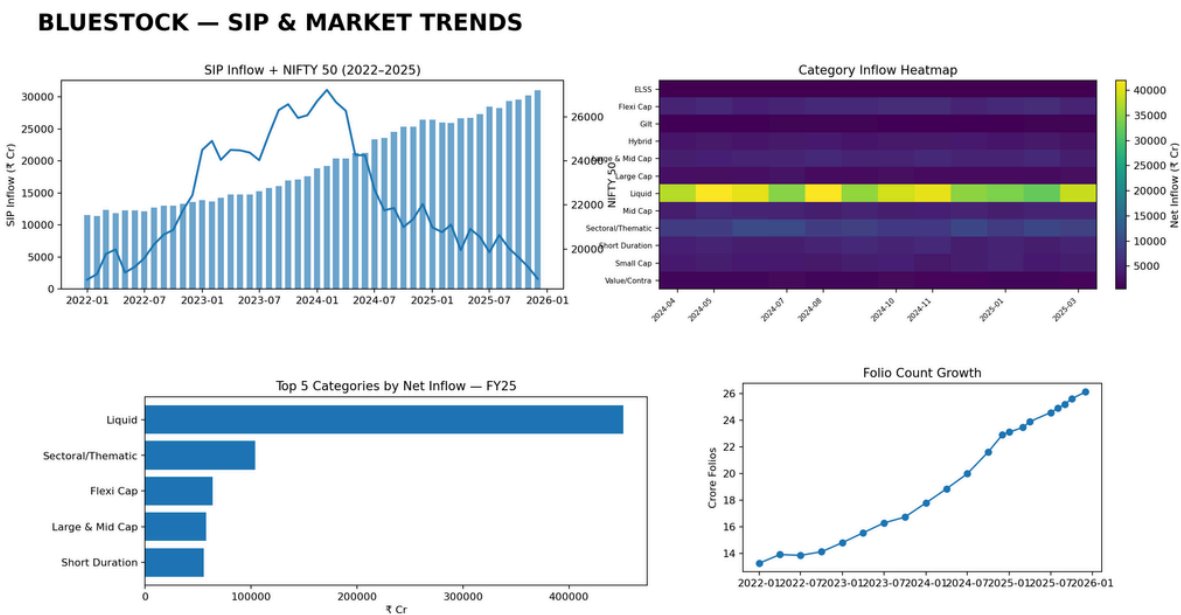
The investor page combines state-level transaction amount, transaction-type mix, average SIP amount by age group, monthly transaction volume and T30/B30 context. State, age-group and city-tier slicers support interactive filtering.

BLUESTOCK — INVESTOR ANALYTICS



14. Dashboard — SIP and Market Trends

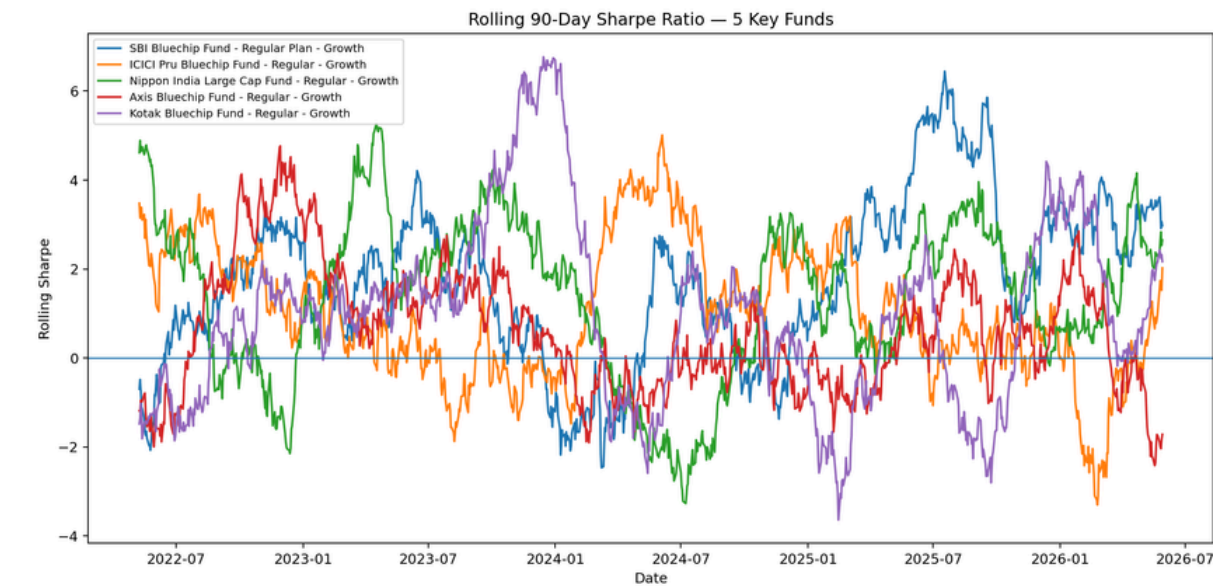
The SIP and market page combines monthly SIP inflow with NIFTY 50, category inflow heatmap, FY25 top-five categories and folio growth. FY25 is defined as April 2024 through March 2025.



15. Advanced Analytics — VaR and Rolling Sharpe

Historical 95% VaR is the 5th percentile of daily returns; CVaR is the mean of returns at or below that threshold. The most negative VaR values in the supplied data occur among small-cap funds.

The rolling 90-day Sharpe analysis is computed for five key schemes: SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip and Kotak Bluechip.



16. Advanced Analytics — Investor Cohorts, Continuity and HHI

The investor transaction window produces cohorts beginning in 2024 and 2025. Among investors with at least six SIP transactions, 1,362 investors qualify for continuity analysis and 1,361 have at least one gap longer than 35 days.

Sector HHI is calculated as the sum of squared sector weights for equity funds. Higher HHI indicates greater concentration; it is used as a risk-context metric rather than as a standalone quality score.

17. Findings, Recommendations and Limitations

Findings: SIP inflows reach the supplied ₹31,002 crore December 2025 peak; folios increase materially across the supplied period; and the composite scorecard favors funds that combine strong three-year performance with risk-adjusted returns, Alpha, lower expense ratios and controlled drawdowns.

Recommendations: Use the dashboard for interactive fund screening; use the scorecard together with drawdown and benchmark context; monitor SIP continuity for investors with long gaps; and use HHI to identify concentrated equity portfolios.

Limitations: The project dataset is a supplied analytical sample rather than a complete live industry database. Some industry KPI values are assignment reference values. The available NAV period does not provide a full five years through the end date. Advanced cohort and continuity results are constrained by the transaction window and should not be generalized beyond the supplied sample.

18. Deliverables and Reproducibility

The project package contains the processed datasets, ETL scripts, SQLite database/schema, EDA notebook and charts, performance notebook and CSV outputs, and the advanced analytics package generated alongside this report.

The native Power BI .pbix remains a desktop artifact: the dashboard must be saved/exported from Power BI Desktop after the required interactive edits. The static dashboard pages in this package are reference/export-ready previews and should not be represented as a native PBIX export.